

KEY TRENDS & RESEARCH PIPELINES

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1. TOP INDUSTRY TRENDS

Trend 1: On-Device AI for XR

The push to run AI models directly on XR hardware, without cloud dependency, is accelerating rapidly. This is driven by the need for low-latency spatial understanding, privacy concerns with streaming camera feeds to the cloud, and the desire for always-available AI even without internet connectivity.

Development	Company	Date	Significance
Snapdragon XR2+ Gen 2 with dedicated NPU (Neural Processing Unit) delivering 12 TOPS	Qualcomm	Oct 2024	Purpose-built AI acceleration for XR headsets enables on-device object detection, hand tracking, and scene understanding simultaneously
Meta Quest 3S ships with on-device hand tracking, scene understanding, and voice commands, all running locally	Meta	Oct 2024	Proves viable consumer-grade on-device AI at \$299 price point
Apple Intelligence integration into visionOS 2.2, on-device LLM for spatial context	Apple	Dec 2024	First integration of an on-device LLM with spatial computing OS
MediaPipe on-device holistic tracking updated with 3D pose, hand, and face simultaneously	Google	Nov 2024	Open-source on-device body understanding pipeline, directly relevant to our work
Qualcomm announces Snapdragon XR Gen 3 targeting 45 TOPS NPU performance	Qualcomm	Feb 2025	4x improvement over Gen 2 will enable much more complex on-device AI by late 2025

- On-device NPU performance is doubling roughly every 12–18 months
- By 2027–2028, headsets will have enough compute to run small LLMs (1B - 7B parameter) on-device
- Hybrid architectures (on-device for latency-critical tasks + cloud for heavy reasoning) will be the dominant pattern
- Apple and Qualcomm are investing heavily in model compression and quantization for on-device deployment

Trend 2: Spatial Computing Convergence

The distinction between AR and VR is dissolving. Every major headset now supports passthrough mixed reality, and the industry is converging on "spatial computing" as the unified paradigm. The device adapts to the user's context, fully immersive when needed, blended with reality when preferred.

Development	Company	Date	Significance
visionOS 2.0 adds volumetric app support and enhanced passthrough	Apple	Sep 2024	Apple doubles down on MR as the default mode, not full VR
Meta Horizon OS opens to third-party hardware manufacturers	Meta	Oct 2024	Creating an "Android of XR", spatial computing becomes a platform, not a device
Android XR announced as Google's spatial computing OS, partnering with Samsung (Project Moohan)	Google + Samsung	Dec 2024	Google enters spatial computing directly, validating the category and fragmenting the market further
Quest 3S delivers color passthrough MR at \$299	Meta	Oct 2024	MR goes mainstream at an accessible price point
Qualcomm Spaces platform updated with unified MR development tools	Qualcomm	Nov 2024	Chip-level push for MR as the default development target

- By 2027, there will be no "VR-only" major headset. Every device will be MR-capable
- Three spatial computing OS ecosystems will compete: visionOS (Apple), Horizon OS (Meta), Android XR (Google/Samsung)
- The winning apps will be those that seamlessly blend digital content with the physical world
- "Spatial computing" will replace "AR/VR" as the industry term within 2 years

Trend 3: GenAI for 3D Content Creation

Generative AI is beginning to solve the biggest bottleneck in XR, content creation. Creating 3D assets, environments, and experiences traditionally requires skilled artists and weeks of work. AI is compressing this from weeks to minutes.

Development	Company	Date	Significance
Meshy v3 Text/image to production-ready 3D models with PBR textures in under 60 seconds	Meshy	Nov 2024	Approaching production quality for simple/medium-complexity assets
Rodin Gen-2 High-fidelity 3D character generation with rigging	Microsoft Research	Jan 2025	Characters come out pre-rigged and animation-ready which is a massive workflow acceleration
Genie 2 Generates playable 3D worlds from a single image	Google DeepMind	Dec 2024	Demonstrates that entire interactive environments can be AI-generated, not just static assets
World Labs (Fei-Fei Li's startup) Raises \$230M for "Large World Models"	World Labs	Nov 2024	Bet on AI that understands and generates entire 3D worlds, not just objects
Trellis by Microsoft Open source 3D asset generation with structured latent representations	Microsoft Research	Dec 2024	Open-source alternative catching up to commercial solutions
3D Gaussian Splatting real-time editing tools emerge (GaussianEditor, DreamGaussian)	Academic Labs	Oct 2024–Feb 2025	Moving from "capture 3D scenes" to "edit and create 3D scenes" with Gaussians
Meta 3D Gen Text to 3D asset in under 1 minute with textures and PBR materials	Meta	Oct 2024	Meta investing in AI-generated 3D content for their XR ecosystem
InstantMesh Single image to 3D mesh in seconds, open source	Tsinghua University	Nov 2024	Academic work pushing quality boundaries, available for anyone to build on

- Within 12 months, generating a basic 3D scene from a text description will be as easy as generating an image today
- AI-generated 3D content will shift from "demo quality" to "production quality" in 2025–2026
- The workflow will evolve: AI generates first draft → human refines → AI iterates. This will be a collaborative loop
- Real-time generation during XR experiences (generate content as the user needs it) is 2–3 years away but being actively researched

Trend 4: Passthrough MR & Environmental Understanding

Passthrough quality is improving dramatically, and devices are moving beyond simple camera feeds to actually understanding the environment, identifying objects, surfaces, room layouts, and spatial relationships.

Development	Company	Date	Significance
Quest 3S Scene API Returns semantic labels (wall, floor, ceiling, couch, table, door, window) with 3D mesh	Meta	Oct 2024	Device-level semantic scene understanding available to all developers
Apple RoomPlan API v2 Real-time room scanning with furniture recognition and parametric 3D models	Apple	Sep 2024	Scans a room and returns a structured 3D model with identified objects
Google Scene Semantics API in ARCore Outdoor scene understanding with terrain, sky, buildings classification	Google	Nov 2024	Extends scene understanding beyond indoor rooms to outdoor environments
Niantic Visual Positioning System (VPS) Centimeter-accurate outdoor localization in 100+ cities	Niantic	Dec 2024	Enables persistent AR content anchored to real-world locations at city scale
LiDAR-quality depth estimation from monocular cameras using AI (Depth Anything V2, Metric3D v2)	Academic Labs	Jan 2025	LiDAR-like depth maps from regular phone cameras democratizes spatial understanding

- Within 2 years, every major XR device will provide a real-time semantic 3D map of the user's environment
- Object-level understanding (not just "there's a surface" but "that's a Samsung refrigerator, model RS27T5200SR") is emerging
- Persistent world maps (your device remembers your home, office, and frequent locations) will become standard
- The gap between what a \$299 Quest sees and what a \$3,499 Vision Pro sees is narrowing rapidly

Trend 5: Voice-First & Multimodal Interfaces for XR

Voice is becoming the primary interaction modality for XR, replacing or augmenting hand controllers. The convergence of real-time voice AI (GPT-4o, Gemini Live) with spatial computing is creating a new interaction paradigm. You talk to your environment and it talks back.

Development	Company	Date	Significance
Meta AI on Ray-Ban "Hey Meta, what am I looking at?" with camera + voice	Meta	Oct 2024	First mass-market product combining voice AI + camera understanding in glasses form factor
GPT-4o real-time voice API Speech-to-speech with <500ms latency, emotion-aware	OpenAI	Oct 2024	Enables natural conversational AI fast enough for real-time spatial interaction
Gemini Live Real-time multimodal conversation with video understanding	Google	Dec 2024	Can see through your camera and discuss what it sees in real-time
Sesame CSM (Conversational Speech Model) Voice AI with personality and emotional nuance	Sesame AI	Jan 2025	Demonstrates that voice AI can feel natural and expressive, not robotic
Moshi by Kyutai Fully open-source speech-to-speech model with <200ms latency	Kyutai Labs	Sep 2024	Open-source alternative enables custom deployment without API dependency
Apple Vision Pro eye tracking + hand tracking + voice as unified input	Apple	Cont.	Demonstrates multimodal (gaze + pinch + voice) as the spatial computing input paradigm

- Voice will be the default interaction mode for spatial computing within 3 years
- Multimodal input (voice + gaze + gesture simultaneously) will become the standard UX pattern
- "Show and ask" (point your camera/gaze at something and ask about it) will be a universal interaction pattern
- Sub-300ms speech-to-speech latency will be standard by 2026, enabling truly conversational spatial AI

Trend 6: AI Agents & Agentic AI in Spatial Contexts

AI is evolving from reactive chatbots to proactive agents that can perceive, reason, plan, and act. When combined with spatial computing, this creates AI that doesn't just answer questions about the world but actively operates within it.

Development	Company	Date	Significance
Google Project Astra AI agent that sees through your camera, understands context, remembers previous interactions, and provides spatial guidance	Google DeepMind	Dec 2024	Closest public demo to a "Spatial AI Assistant". Directly validates our product vector
Meta FAIR Ego4D + Ego-Exo4D datasets Massive first-person video datasets for training embodied AI	Meta FAIR	Cont.	Training data specifically designed for AI that understands the world from a human's perspective
OpenAI Operator AI agent that can take actions in digital environments	OpenAI	Jan 2025	While not spatial yet, demonstrates the "agent that acts" paradigm moving mainstream
NVIDIA GR00T Foundation model for humanoid robot understanding of 3D environments	NVIDIA	Oct 2024	Spatial understanding + action in the physical world. Robotics and XR are converging
SpatialBot LLM fine-tuned for spatial reasoning, understanding relative positions, and giving spatial instructions	Academic (multiple labs)	Feb 2025	Research specifically on making LLMs understand "left of," "behind," "two steps forward"

- AI agents that understand 3D space will be a major research frontier through 2025–2027
- The line between "XR assistant" and "embodied AI agent" will blur
- Proactive spatial AI (the AI notices things and alerts you, rather than waiting to be asked) is an emerging paradigm
- Memory and persistence (the AI remembers your spaces, your preferences, your routines) will be a key differentiator

Trend 7: Smart Glasses as the Next Computing Platform

The industry consensus is shifting: smart glasses (not headsets) will be the eventual mass-market spatial computing form factor. Meta's Ray-Ban success, combined with multiple new entrants, is accelerating this timeline.

Development	Company	Date	Significance
Meta Ray-Ban smart glasses sell 2M+ units in first year, outpacing all expectations	Meta	Oct 2024	Proves consumer demand for AI-enabled glasses, even without AR display
Meta Orion AR glasses prototype with full holographic display demonstrated	Meta	Sep 2024	Most advanced AR glasses ever publicly shown. Neural interface wristband for control
Snap Spectacles 5th Gen AR glasses with Snap OS for developers	Snap	Sep 2024	Lightweight AR glasses with spatial understanding, developer-focused
Xreal Air 2 Ultra Consumer AR glasses with 6DoF tracking and spatial computing	Xreal	Cont.	Sub-\$500 AR glasses with real spatial capabilities
Brilliant Labs Frame Open-source AI glasses with Gemini/GPT integration	Brilliant Labs	Oct 2024	Fully hackable AI glasses for \$349 Developer-friendly
Google + Samsung Android XR glasses announced for 2025 release	Google + Samsung	Dec 2024	Google re-entering glasses with Samsung hardware partnership
Apple reportedly working on lightweight AR glasses (post-Vision Pro)	Apple (reports)	2024-25	Apple's long-term play is glasses, not headset

- By 2027, multiple companies will ship consumer AR glasses with spatial displays under \$500
- Meta's Ray-Ban line proves the stepping-stone strategy: AI-first glasses (no display) → add display later
- The first "iPhone moment" for glasses is expected 2027–2029
- Glasses will be the primary platform for spatial AI assistants

Trend 8: Real-Time 3D Reconstruction & Neural Scene Representations

The ability to create high-quality 3D representations of real environments in real time, using techniques like 3D Gaussian Splatting and Neural Radiance Fields, has improved dramatically. This is moving from research demos to practical applications.

Development	Company	Date	Significance
3D Gaussian Splatting achieves real-time rendering on mobile devices (30fps on iPhone 15)	Multiple academic labs	Jan 2025	3DGS moves from desktop-only to mobile
Nerfstudio 1.0 Production-ready framework for training and deploying neural scene representations	UC Berkeley	Oct 2024	NeRF/3DGS tooling matures from research code to production framework
Luma AI releases real-time Gaussian Splatting viewer and capture app on iOS	Luma AI	Nov 2024	Consumer-facing app for capturing and sharing 3D scenes Proves market readiness
SplaTAM Simultaneous tracking and mapping using Gaussian Splatting in real-time	CMU	Oct 2024	Combining SLAM with Gaussian Splatting 3D map of your world as you move
GaussianEditor Edit real-world 3D scenes captured with Gaussian Splatting using text prompts	Academic	Nov 2024	"Remove the chair" or "change the wall color" in a 3D capture of a real room
Meta publishes research on instant 3D scene generation from text for VR environments	Meta FAIR	Dec 2024	Connecting GenAI with scene reconstruction. Generates entire rooms from descriptions

- Real-time 3D reconstruction on mobile devices will be standard within 18 months
- 3D Gaussian Splatting is emerging as the preferred representation over NeRFs for real-time applications
- The pipeline will become: scan → reconstruct → understand → augment → interact
- This enables "digital twins on the fly". Point your phone at a room, get a full 3D model instantly

Trend 9: Privacy-First Spatial Computing

Foundation model companies (OpenAI, Google, Meta, Anthropic) are actively working on making their models understand the physical world, not just text and images, but 3D space, physics, and embodied interaction.

Development	Company	Date	Significance
GPT-4o demonstrates spatial reasoning and can describe relative positions of objects, understand room layouts from images	OpenAI	Dec 2024	Foundation models are gaining basic spatial intelligence
Gemini 2.0 with native tool use and spatial understanding capabilities	Google	Dec 2024	Google explicitly positioning Gemini as an "agentic" model that understands physical context
Meta FAIR "3D-LLM: Injecting 3D World into Large Language Models" research	Meta FAIR	Nov 2024	Directly addresses giving LLMs 3D scene understanding
SpatialVLM Vision-language model fine-tuned for spatial relationships and measurements	Google Research	Dec 2024	Model that can estimate distances, understand "left/right/behind" from images
PhysGen AI models that understand physics for generating realistic interactions	Academic (MIT, Stanford)	Jan 2025	Toward AI that understands not just what objects are but how they behave
EmbodiedGPT Connecting LLMs to physical world perception and action	Academic (Multiple)	Feb 2025	Active research area with dozens of papers in last 6 months

- Within 2 years, major LLMs will natively understand 3D space from images/video without special fine-tuning
- "Spatial reasoning" will become a standard benchmark for foundation models (alongside math, coding, language)
- The gap between "AI that describes a photo" and "AI that understands a 3D scene" will close rapidly
- This is possibly the single most important trend for our product vector

2. MAJOR RESEARCH PIPELINES

Meta (Reality Labs + FAIR)

Meta is our most important competitor to study. Their Ray-Ban AI is a very strong product, but it lacks deep spatial understanding. Their research pipeline (Ego4D, Scene API, Codec Avatars) is building the foundations we would also need.

Research Area	Key Projects	Status	Relevance
Codec Avatars	Photorealistic real-time avatars from headset sensors. Latest: "Universal Codec Avatars" Generate avatar from a phone scan	Active research, demos shown Sep 2024	Relevant to our character generation pillar
Ego4D / Ego-Exo4D	Massive first-person video datasets for training embodied AI 4,000+ hours of daily-life video with annotations	Ongoing, new data releases Nov 2024	Training data for AI that understands the world from human POV
Scene Understanding	Quest Scene API, room mesh, semantic labels, object recognition Continuously improving	Shipping in Quest 3/3S	Core to our Spatial AI Assistant
Project Aria	Research glasses with sensors for spatial AI research Collecting data for future AR glasses	Active research device	Informs what sensors future glasses will have
Meta 3D Gen	Text-to-3D asset generation with PBR materials in under 1 minute	Research paper Oct 2024	Relevant to our GenAI for 3D content pillar
Orion AR Glasses	Most advanced AR glasses prototype Holographic waveguide display, EMG wristband	Prototype shown Sep 2024, consumer version reportedly 2027	Defines what future glasses hardware will look like
Meta AI Assistant	Voice + camera AI on Ray-Ban glasses "Hey Meta, what am I looking at?"	Shipping product, continuously updated	Closest existing competitor to our Spatial AI Assistant on glasses
Acoustic Simulation	Realistic spatial audio based on room geometry (sounds interact with virtual and physical objects)	Research, partially shipping	Important for immersive spatial AI interactions

Google (DeepMind + Google Research + ARCore)

Google's Project Astra is the single most important competitor/reference for our Spatial AI Assistant. We need to study it deeply, understand what it does well, what it can't do, and where the gaps are. Their research pipeline is building toward exactly the same vision.

Research Area	Key Projects	Status	Relevance
Project Astra	Real-time AI agent that sees through camera, understands spatial context, remembers objects and locations, provides guidance	Demo updated Dec 2024, rumored integration into Gemini 2025	Most directly competitive to our Spatial AI Assistant vision
Gemini 2.0	Multimodal foundation model with native spatial understanding, tool use, and agentic capabilities	Launched Dec 2024	Potential foundation model for our AI assistant
SpatialVLM	Vision-language model specifically for spatial relationships, distance estimation, relative positioning	Research paper Dec 2024	Directly relevant to spatial reasoning in our assistant
Android XR	Spatial computing OS for headsets and glasses Partnering with Samsung, Qualcomm	Announced Dec 2024, devices expected 2025	New platform target for our solution
ARCore Geospatial API	Cloud anchors, street-level localization, persistent AR at city scale	Shipping, updated Dec 2024	Relevant for outdoor Spatial AI Assistant use cases
Genie 2	Generates playable 3D worlds from a single image Interactive, explorable environments	Research demo Dec 2024	Relevant to our generative world building research
Scene Semantics API	Outdoor scene understanding Terrain, buildings, sky, vegetation classification	Shipping in ARCore	Extends scene understanding to outdoor environments

Apple (Spatial Computing + Apple Intelligence)

Apple's approach is privacy-first, on-device-heavy, and premium.

Their APIs (RoomPlan, ARKit) are among the best for spatial understanding. Their limitation: Siri/Apple Intelligence is weaker than GPT-4o/Gemini for conversational AI, which is a gap we could exploit on their platform.

Research Area	Key Projects	Status	Relevance
visionOS + Apple Intelligence	On-device LLM integration with spatial computing Siri gains contextual awareness of the physical space	visionOS 2.2, Dec 2024	Apple's approach to spatial AI is privacy-first, on-device
RoomPlan API v2	Real-time room scanning with furniture/object recognition, returns parametric 3D models	Shipping, updated Sep 2024	One of the best room understanding APIs available. Can be a benchmark for us
ARKit 7	Updated plane detection, scene reconstruction mesh, persistent anchors, object tracking	Shipping	Foundation for iOS AR development
Object Capture API	Photogrammetry-based 3D scanning from iPhone photos Create 3D models of real objects	Shipping, improved with LiDAR devices	Relevant to 3D content pipeline
Spatial Personas	Photorealistic avatars for FaceTime on Vision Pro Shared spatial experiences	Shipping	Relevant to our character/avatar research
Lightweight AR Glasses (rumored)	Post-Vision Pro consumer glasses Reportedly targeting 2027-2028	Rumored, multiple credible reports	Long-term platform target
Core ML + Neural Engine	On-device ML framework optimized for Apple silicon 18 TOPS on M2 chip	Shipping	Reference for on-device AI capabilities

NVIDIA

Research Area	Key Projects	Status	Relevance
GR00T (Generalist Robot 00 Technology)	Foundation model for humanoid robots Understanding 3D environments and taking physical actions	Research, Oct 2024	Spatial understanding for robotics overlaps with spatial AI for XR
Omniverse / Digital Twins	Platform for creating and simulating 3D digital twins of real environments	Shipping, enterprise-focused	Relevant to our 3D scene reconstruction and persistent world research
CloudXR	Cloud-based rendering and streaming for XR High-fidelity visuals streamed to lightweight devices	Shipping	Enables our solution to run heavy rendering in cloud when needed
Isaac Sim	Robotics simulation with photorealistic environments and physics	Shipping	Training environment for spatial AI agents
NeMo + TensorRT-LLM	LLM optimization and deployment Quantization, pruning for on-device deployment	Shipping	Relevant to deploying AI models on XR hardware

Academic Labs & Key Research Groups

Lab / Group	Institution	Focus Area	Key Recent Work
Robotics & Embodied AI Lab	Stanford	Embodied AI, spatial reasoning	EmbodiedGPT, SayCan LLMs that understand and act in physical spaces
BAIR (Berkeley AI Research)	UC Berkeley	3DGS, NeRF, scene understanding	Nerfstudio 1.0, SplaTAM, real-time 3D reconstruction
Visual Computing Lab	MIT CSAIL	Scene understanding, spatial AI	ConceptFusion (open-vocabulary 3D scene understanding), physics-aware AI
Ego-Centric Perception Group	Meta FAIR + Universities	First-person AI understanding	Ego4D benchmark, egocentric action recognition
3D Vision Lab	ETH Zurich	3D reconstruction, SLAM	Advanced SLAM systems, large-scale 3D mapping
Graphics & AI Lab	Tsinghua University	3D generation, Gaussian Splatting	InstantMesh, GaussianEditor, 3DGS advances
Spatial Intelligence Lab	Google Research	Spatial VLMs, scene graphs	SpatialVLM, 3D scene graph generation
Mixed Reality & AI Lab	CMU	AR/VR systems, spatial AI	SplaTAM, real-time spatial understanding systems
HCI + XR Lab	Georgia Tech	Interaction paradigms, multimodal	Research on voice + gesture + gaze multimodal interfaces

3. KEY PAPERS

Paper	Authors / Lab	Key Contribution
"SpatialVLM: Endowing Vision-Language Models with Spatial Reasoning"	Google Research	Fine-tuning VLMs to understand spatial relationships, distances, and relative positions
"3D-LLM: Injecting the 3D World into Large Language Models"	Meta FAIR + UCLA	Method to give LLMs 3D scene understanding from point clouds
"SplaTAM: Splat, Track & Map 3D Gaussians for Dense RGB-D SLAM"	CMU	Real-time SLAM using 3D Gaussian Splatting 3D map of the world as you move
"ConceptFusion: Open-set Multimodal 3D Mapping"	MIT + Google	Create 3D maps where every point has semantic meaning
"LEO: An Embodied Generalist Agent in 3D World"	Multiple Universities	AI agent that can perceive, ground, reason, plan, and act in 3D environments
"GaussianEditor: Swift and Controllable 3D Editing with Gaussian Splatting"	Multiple Universities	Edit real-world 3D captures using text prompts
"Depth Anything V2"	TikTok Research + HKU	State-of-the-art monocular depth estimation (LiDAR-quality depth from single camera)
"EmbodiedScan: A Holistic Multi-Modal 3D Perception Suite"	Multiple Universities	Benchmark for AI understanding of 3D environments from egocentric views
"SceneScript: Reconstructing Scenes With An Autoregressive Structured Language Model"	Meta FAIR	Reconstructing 3D room layouts using a language model architecture
"WorldGPT: A Generative World Model for Physical Reasoning"	Academic	LLM that can predict physical outcomes in 3D environments